

Collective dynamics in binary superfluids: From dissipationless flow to dispersive shock waves

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anr®

1. Introduction

Binary superfluids...

- Condensates of interacting ^{atoms} exciton polaritons in two spin states /

Proc. Int. Sch. Phys. Enrico Fermi **211** (2025)

Rev. Mod. Phys. **85**, 299 (2013)

Elliptically polarized laser in a nonlinear medium / ...

Adv. At. Mol. Opt. Phys. **74**, 157 (2025)

- $$H = \int d^d \mathbf{r} \left(\frac{|\nabla \psi_+|^2 + |\nabla \psi_-|^2}{2} + \underbrace{\frac{|\psi_+|^4 + |\psi_-|^4}{2}}_{\text{Intra-component interactions}} + \underbrace{\alpha |\psi_+|^2 |\psi_-|^2}_{\text{Inter-component interactions}} \right) \quad \alpha = \frac{g_{+-}}{g_{\pm\pm}}$$

- $$\psi_{\pm} = \sqrt{\rho_{\pm}} \exp\left(i \int \mathbf{v}_{\pm} \cdot d\mathbf{r}\right)$$

Density channel	$\begin{cases} \rho = \rho_+ + \rho_- \\ \mathbf{v} = \frac{\mathbf{v}_+ + \mathbf{v}_-}{2} \end{cases}$	Spin channel	$\begin{cases} \sigma = \rho_+ - \rho_- \\ \mathbf{v} = \mathbf{v}_+ - \mathbf{v}_- \end{cases}$
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- Miscibility problems / Collective and topological excitations / ...

...of matter

³⁹K condensate in two Rabi-coupled Zeeman states



Phys. Rev. Lett. **128**, 083401 (2022)

Thomas Bourdel's group
LCF, Palaiseau

...of light

Elliptically polarized laser in a warm ⁸⁷Rb vapor



Phys. Rev. Lett. **134**, 223403 (2025)

Quentin Glorieux's group
LKB, Paris

2. Critical speed of a binary superfluid of light

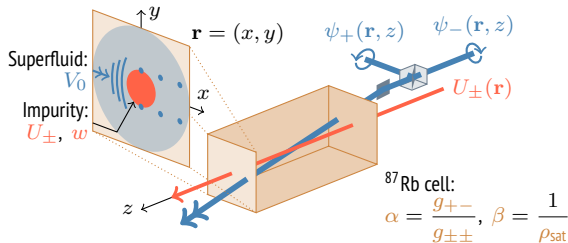
Eur. Phys. J. D **80**, 72 (2026)



Claire Michel
INPHYNI, Nice



Nicolas Cherroret
LKB, Paris



Model equations

- Two-component inhomogeneous nonlinear Schrödinger-type equation, analogous to the Gross-Pitaevskii equation of Bose-Bose superfluid mixtures:

$$i\partial_z \psi_{\pm} = \left[-\frac{\nabla^2}{2} + U_{\pm}(\mathbf{r}) + \frac{|\psi_{\pm}|^2 + \alpha|\psi_{\mp}|^2}{1 + \beta(|\psi_+|^2 + |\psi_-|^2)} \right] \psi_{\pm}$$

- What condition must the incident velocity V_0 satisfy to ensure superfluid, dissipationless flow?

$$(\psi_{\pm})_{\infty} = \underbrace{\sqrt{\frac{1}{2}} \exp[i(V_0 x - \mu z)]}_{\text{Fully balanced}}$$

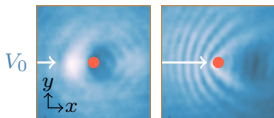
$$\underbrace{U_{\pm} > 0}_{\text{Repulsive}}$$

$$\underbrace{g_{\pm\pm} > g_{+-} > 0}_{\text{Miscible \& repulsive}} \implies 0 < \alpha < 1$$

Linear response

- Impurity $\ll$ Interactions
- Impurity-induced excitations:
Density and spin Bogoliubov waves

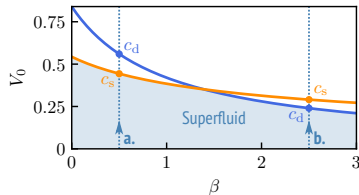
Single-component superfluid of light
in a photorefractive crystal



Nat. Commun. 9, 2108 (2018)

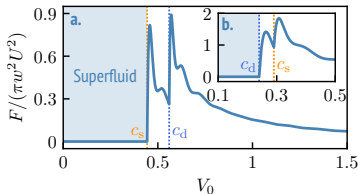
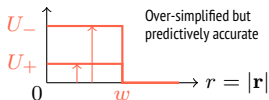
Landau criterion

$$V_0 < V_c = \min \left\{ c_d = \sqrt{\frac{1+\alpha}{2}} \frac{1}{1+\beta}, c_s = \sqrt{\frac{1-\alpha}{2}} \frac{1}{\sqrt{1+\beta}} \right\}$$



Drag force

$$\mathbf{F} = \int d^2\mathbf{r} \begin{bmatrix} \psi_+^* & \psi_-^* \end{bmatrix} \begin{bmatrix} \nabla U_+(\mathbf{r}) & 0 \\ 0 & \nabla U_-(\mathbf{r}) \end{bmatrix} \begin{bmatrix} \psi_+ \\ \psi_- \end{bmatrix}$$



Superfluid hydrodynamics

- Stationary flow with asymptotically uniform densities and velocities: $(\rho, \sigma, \mathbf{V}, \mathbf{v})_\infty = (1, 0, V_0 \hat{\mathbf{x}}, \mathbf{0})$

- Bernoulli equations: $\cos \theta = \sigma / \rho \quad U, u = U_+ \pm U_-$

$$\begin{cases} \frac{(1 + \alpha)\rho}{1 + \beta\rho} = \frac{1 + \alpha}{1 + \beta} - U \mathbf{1}_{r < w} - (V^2 - V_0^2) - \frac{v^2}{4} + \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} + \frac{\nabla \cdot (\rho \nabla \theta)}{2\rho \tan \theta} - \frac{(\nabla \theta)^2}{4} \\ \frac{(1 - \alpha)\sigma}{1 + \beta\rho} = -u \mathbf{1}_{r < w} - \mathbf{V} \cdot \mathbf{v} - \frac{\nabla \cdot (\rho \nabla \theta)}{2\rho \sin \theta} \end{cases}$$

- Continuity equations:

$$\nabla \cdot \left(\rho \mathbf{V} + \frac{\sigma \mathbf{v}}{2} \right) = 0 = \nabla \cdot \left(\sigma \mathbf{V} + \frac{\rho \mathbf{v}}{2} \right)$$

Elliptic flow

The superflow monotonically flattens at infinity $\iff$ The continuity equations are strongly elliptic

Chaplygin method

Sci. Mem. Moscow Univ. Math. Phys. **21**, 1 (1902)

Physica D **148**, 221 (2001) $\longrightarrow$ Impenetrable disk

Phys. Rev. A **109**, 013317 (2024) $\longrightarrow$ Disk of arbitrary penetrability

Local-field generalization of the Landau criterion
for a fully imbalanced mixture

$$\text{StrEll}[\underbrace{\rho(\mathbf{V}, \mathbf{v})}_{\text{Bernoulli}}, \underbrace{\sigma(\mathbf{V}, \mathbf{v})}_{\text{Bernoulli}}, \mathbf{V}, \mathbf{v}] > 0$$

$$\text{StrEll}(\mathbf{r}; V_0, \alpha, \beta, U, u) > 0, \quad \forall \mathbf{r} = (x, y)$$

$$\iff V_0 < V_c(\alpha, \beta, U, u)$$

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- Continuity equations: Compressibility $\ll 1$

$$\nabla \cdot \left(\rho \mathbf{V} + \frac{\sigma \mathbf{v}}{2} \right) = 0 = \nabla \cdot \left(\sigma \mathbf{V} + \frac{\rho \mathbf{v}}{2} \right) \quad \begin{cases} \nabla \cdot \mathbf{V} = 0 = \nabla \cdot \mathbf{v} \\ \text{Appropriate boundary conditions at } r = w \end{cases}$$

Elliptic flow

The superflow monotonically flattens at infinity $\iff$ The continuity equations are **strongly elliptic**

Chaplygin method

Sci. Mem. Moscow Univ. Math. Phys. **21**, 1 (1902)

Physica D **148**, 221 (2001) $\longrightarrow$ Impenetrable disk

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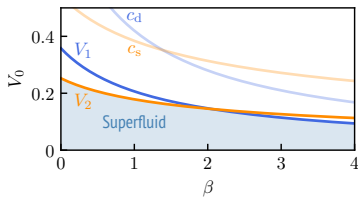
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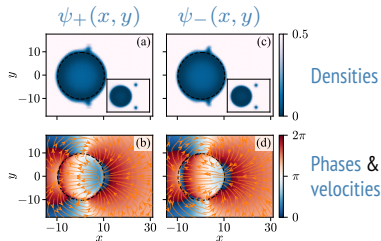
Impenetrable impurity

$$U > (1 + \alpha)/(1 + \beta)$$

$$V_c = \min \left\{ V_1: c_d \sqrt{\frac{(1 + \beta)[\sqrt{(1 + \beta)(121 + 25\beta)} - (11 + 5\beta)]}{9\beta}}, V_2: c_s \sqrt{\frac{2(1 + \alpha)}{11 + 5\alpha}} \right\}$$



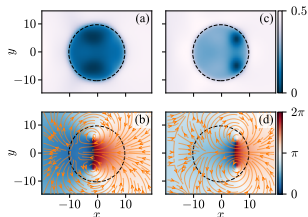
Vortex-antivortex pairs at the impurity poles



Penetrable impurity

Cumbersome yet analytical $V_c(\alpha, \beta, U, u)$

Jones-Roberts solitons inside the impurity



3. Nonlinear periodic waves in a binary Bose-Einstein condensate

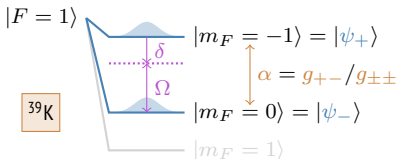
Thibault Congy
Northumbria University, Newcastle



Patrick Sprenger
University of California, Merced



Bose-Einstein condensate in two Rabi-coupled Zeeman states



$$\mathcal{E}(\theta) = -\frac{\Omega \sin \theta - \delta \cos \theta}{2} \rho + \left(1 - \frac{1 - \alpha}{2} \sin^2 \theta\right) \frac{\rho^2}{2}$$

$\rho = \text{const}$

Cubic-quintic nonlinear Schrödinger modeling

Petrov, Orsay / Bourdel, Palaiseau / Tarruell, Barcelona

Ground state for $\frac{|1-\alpha|\rho}{\Omega} \ll 1$:

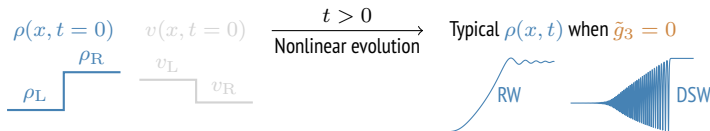
$$\mathcal{E}_{\text{gs}} \simeq -\frac{\sqrt{\Omega^2 + \delta^2}}{2} \rho + g_2(\alpha, \Omega, \delta) \frac{\rho^2}{2} + g_3(\alpha, \Omega, \delta) \frac{\rho^3}{3}$$

$\implies$ Effective cubic-quintic nonlinear Schrödinger dynamics for a scalar field $\varphi(\mathbf{r}, t)$ with density $|\varphi|^2 = \rho$:

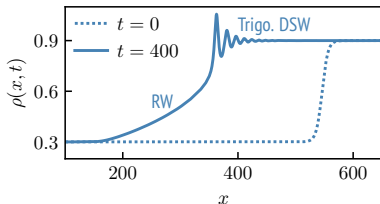
$$i\partial_t \varphi = \left(-\frac{\nabla^2}{2} + g_2 |\varphi|^2 + g_3 |\varphi|^4 \right) \varphi$$

Contact dispersive shock waves

- 1D geometry:
$$i\partial_t \varphi = \left(-\frac{\partial_{xx}}{2} + \tilde{g}_2 |\varphi|^2 + \tilde{g}_3 |\varphi|^4 \right) \varphi \quad \varphi = \sqrt{\rho} \exp\left(i \int v \, dx\right)$$
- Riemann initial-value problem:



- Contact DSWs emerge when $\tilde{g}_3 \neq 0$:



Whitham modulation theory

Nonlinear Periodic Waves and Their Modulations (2000)
 Physica D 333, 11 (2016)

Within experimental reach at LCF:

$$(\Delta x, \Delta t)_{\text{Contact DSW}} \simeq (14.4 \, \mu\text{m}, 318.4 \, \text{ms})$$

$$\begin{aligned} \omega_{\perp} / (2\pi) &= 300 \, \text{Hz} & \Omega / (2\pi) &= 25.4 \, \text{kHz} \\ \rho &\simeq 2.5 \times 10^9 \, \text{m}^{-1} & \delta / \Omega &= 0.9 \end{aligned}$$

4. Self-propulsion of a mobile impurity against a superflow

Phys. Rev. Lett. **136**, 223401 (2026)
arXiv:2609.03948

Myrann Baker-Rasooli
LKB, Paris

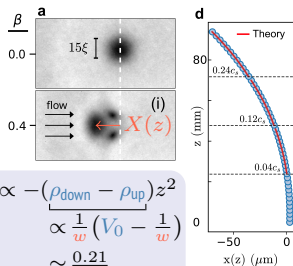


Tangui Aladjidi
LKB, Paris

Upstream motion via downstream vortices

$$i\partial_z\psi = \left[-\frac{\nabla^2}{2} + U(\mathbf{r} - X\hat{\mathbf{x}}) + |\psi|^2 \right] \psi$$

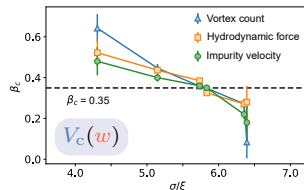
$$\frac{d^2 X}{dz^2} \propto \int d^2\mathbf{r} |\psi|^2 \partial_x U(\mathbf{r} - X\hat{\mathbf{x}})$$



$$\begin{aligned} X(z) &\propto -(\rho_{\text{down}} - \rho_{\text{up}})z^2 \\ &\propto \frac{1}{w} (V_0 - \frac{1}{w}) \\ &\simeq \frac{0.21}{w} \end{aligned}$$

Critical speed for vortex shedding

- Three-observable measurement for a hard-core scatterer of large tunable radius



- Available theory:

$$V_c(w) \simeq 0.36(9) + \frac{0.32(2)}{(2\sqrt{2}w)^{0.61(5)}}$$

Physica D **148**, 221 (2001) Physica D **140**, 126 (2000)

5. Summary and outlook

Critical speed of a binary superfluid of light

Eur. Phys. J. D **80**, 72 (2026)

Special issue "Paraxial Fluids of Light" still open for submissions!

- *Superfluidity threshold*: Analytical derivation of the critical speed from the theory of elliptic flows across the full two-component parameter space
- *Breakdown mechanism*: Driven by the nucleation of vortex-antivortex pairs or Jones-Roberts solitons in both circular polarizations of the optical field

Nonlinear periodic waves in a binary Bose-Einstein condensate

In progress

- *Effective model*: Scalar cubic-quintic nonlinear Schrödinger description for a Bose-Einstein condensate with two coherently coupled Zeeman states
- *Dispersive dynamics*: Unveiling contact dispersive shock waves in a 1D geometry via Whitham modulation theory and numerical simulations

Self-propulsion of a mobile impurity against a superflow

Phys. Rev. Lett. **136**, 223401 (2026)

arXiv:2609.03948

- *Upstream self-propulsion*: Demonstration of an all-optical impurity "swimming" against a 2D flow of superfluid light via vortex-antivortex shedding
- *Critical speed*: Multi-observable benchmark for a hard-core impurity using vortex count, hydrodynamic force, and impurity displacement

Ongoing work and future directions

- **Quantum** depletion of a **3D** superfluid of light $\times$ LKB
- Post-**quench** dynamics of a **disordered** superfluid of light $\times$ INPHYNI
- **KPZ** dynamics in a tower of **quantum many-body scars** $\times$ LPTMS
- Post-**quench** dynamics of a quantum **mixture** $\times$ LCF
- **Rayleigh-Plesset** dynamics in an **immiscible** quantum mixture $\times$ LPTMS

Phys. Rev. A **92**, 043802 (2015)

EPL **132**, 66001 (2021)

Phys. Rev. B **112**, 134314 (2025)

Phys. Rev. Lett. **130**, 153401 (2023)

Merci !